

Alexander Loker

aloker@uwaterloo.ca | linkedin.com/in/alexander-loker | alexloker.com

Education

University of Waterloo

2025 – Present

Bachelor of Applied Science in Mechatronics Engineering

Skills

Hardware: Altium, LTspice, SolidWorks, Raspi, AutoCAD

Software & Protocols: C++, Python, Git, Linux, PWM, DMX512, I2C, SPI, UART

Tools: Oscilloscope, Soldering, 3D Printing

Experience

WET Design

May – August 2026

Product Designer & Engineer

Sun Valley, CA, USA

- Built a handheld DMX light tester from the ground up to replace the **\$1200** Swisson XMT-500, bringing per unit cost to **\$300** in parts while adding fault protection and an externally programmable layer.
- Wired and tested the Waterglo prototype end to end, an illuminated water and air feature requiring lighting and pump control, driving both from a single DMX universe through a WET DMX to power supply converter.
- Owned the team's Weigl ProCommander HXi integration for a fountain rig, wiring and documenting the control box across DMX, analog and digital I/O, and training 4 engineers on the platform.

Projects

DMX Handheld Light Tester – WET Design

June – August 2026

- Designed a custom **Raspi CM4** carrier board in **Altium** from part selection to schematic to 4 layer PCB layout, integrating a WET RS-485 port, battery management, and power distribution.
- Maintained **signal integrity** and reduced **EMI** through guarding differential pairs, a ferrite-bead pi filter, tight local **decoupling** at each IC's power pins, and a minimized switching-current loop to reduce parasitic inductance.
- Integrated battery management and power distribution across a **BQ charger IC**, **buck-boost IC**, and CM4 3.3V rail, using low-impedance routing and thermal balancing to carry up to **10W** safely.
- Wrote the tester software in Python, building **DMX512** framing and timing on top of **UART** alongside **I2C** and **SPI** drivers for the BMS and LCD.
- Iterated the enclosure around software and PCB revisions, ensuring precise and tight port-fits while customizing UX/UI in **SolidWorks** and **PyGame** to accommodate a modular script, work-site safe and one-handed design.

Companion Robot

January – April 2026

- Led end-to-end design of a companion robot integrating mechanical design (**SolidWorks**), electrical components and routing, and firmware implementation (**Python**).
- Designed **hardware and firmware** for a tri-omniwheel drivetrain on a **Pi Pico** with **Python PID** motor control and a tracking algorithm, MediaPipe, enabling **autonomous 360°** human-following.
- Compacted robot volume by **26%** by redesigning the mechanical housing alongside embedding servos, microphones, and a camera module directly into the shell, utilizing compact interlocking joints.

Under-voltage Lockout Circuit – Biomechatronics Exoskeleton Team

February – March 2026

- Designed and verified a UVLO circuit schematic in **LTspice**, and transformed the schematic into a PCB in **Altium**, confirming safe voltage input to sensitive components.
- Created under-voltage lockout logic via **Schmitt trigger hysteresis** and controlled current/voltage flow through a **dual-MOSFET switch system** ensuring voltage lockout at **24.1V**.

Autonomous Air Hockey Defense Machine

November 2025

- Designed and assembled a **fully automated** puck-deflection robot intercepting **76 of 100** shots against live opponents.
- Integrated an **optical sensor** for real-time foul detection and a **distance sensor** for trigonometric puck trajectory prediction, collecting and analyzing sensor data to validate system accuracy.
- Developed a real-time **predictive algorithm in C++** to calculate puck landing position from sensor input.